

Fig. 1: With active PFC, the current consumption is controlled by pulse-width modulation to near-sinusoidal shape

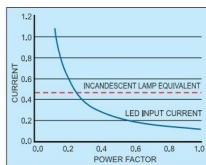


Fig. 2: Comparison of incandescent bulb and LED driver with power factor of less than one

with power factor correction (PFC). EnergyStar is even more explicit, prescribing a power factor of 0.9 or better for commercial drivers.

Without active PFC, it is, however, only possible to reach values that are significantly lower—around 0.5 or even less, depending on the power rating. AC/DC LED drivers therefore need to be equipped with special PFC circuits. Their principle is straight-forward: instead of connecting the charging capacitor directly to the rectifier, a pulse-width modulator is installed between the two components. This modulator ensures that the capacitor is charged by several small-current pulses during the half wave. The current consumption is therefore more or less synchronised with the mains voltage and approximately sinusoidal

(refer Fig. 1).

A well-designed PFC circuit in some LED drivers increases the power factor to a value of around 0.95, thus exceeding the stringent EnergyStar requirements as well as EN 61000 specifications. Although it is technically possible to achieve even better values, the associated costs outweigh the benefits.

While EN 61000-3-2 requires a power factor of greater than 0.9 only from 25W upwards, active PFC also makes sense at lower power rates. This becomes obvious if one considers that many circuits include a large number of small- or medium-power LED luminaires or consist of small luminaire clusters with separate drivers. Since ten 12W loads consume a total of 120W, mains network operators would probably appreciate it if a proper power factor correction was applied. This is why some manufacturers offer products with active PFC from as low as 12W.

Relationship between PFC and efficiency

Many people incorrectly believe that a driver with a low power factor offers poor efficiency. While such drivers consume considerably more energy from the mains than is required for

powering the LEDs, a large share of this power is actually fed back to the mains network. This share is thus not lost, as would be the case with a low-efficiency device. It is simply fed from the 'wrong' side. This is probably the reason why many people confuse the power factor value with efficiency.

Fig. 2 compares the current consumption of a 100W incandescent lamp (red curve) with that of a 25W LED. Both devices produce about the same amount of light. The incandescent bulb with a power factor of one consumes a constant current of 0.45A from the 230V mains network. With an LED driver of power factor one, the current consumption would be around 0.11A. At a power factor of 0.95, it would be slightly higher. At a power factor of 0.25, the current consumption would amount to 0.45A, which corresponds to that of the incandescent lamp—the actual LED output would, however, only be 25W. The remaining 75W is returned through the 'wrong' phase back to the mains. The energy is thus not lost and the reactive current is not metered by the power meter.

For AC/DC drivers, active power factor correction is, however, as important as high efficiency, especially if one takes into account that billions of such drivers will be connected to the mains over the next few years. Power factor correction is therefore not so much geared towards keeping electricity costs down, but helps maintain the quality of the mains power by eliminating harmonic interference.

Rapid pace of development in the field of power LEDs and drivers

For the foreseeable future, the industry is focusing on solutions for the existing infrastructure in residential and office buildings. For LED luminaires, this means that they must thus be dimmable with conventional TRIACs. This creates a number of technical problems, since the leading or trailing edge control of dimmers and PFC circuits of drivers interfere with each other. Conventional drivers can therefore not be